CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 6: Solving Problems Based on Quadratic Equations

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find two consecutive positive odd numbers whose product is 143.

2. The sum of the squares of two consecutive natural numbers is 113. Find the numbers.

3. A positive number exceeds its reciprocal by $\frac{15}{4}$. Find the number.

4. The product of Meera's age 2 years ago and her age 4 years from now is 91. Find her present age.

5. The length of a rectangle is 5 cm more than its breadth and its area is 84 cm². Find its dimensions.

6. The sides of a right-angled triangle containing the right angle are x cm and $(x + 3)$ cm, and its area is 54 cm². Find the sides.

7. A train covers 300 km at a uniform speed. If the speed had been 15 km/h more, the journey would have taken 1 hour less. Find the speed.

8. A boat takes 1 hour more to go 12 km upstream than to return downstream. If the speed of the stream is 1 km/h, find the speed of the boat in still water.

9. Some students planned a picnic costing ₹480 in all. When 4 students dropped out, each remaining student had to pay ₹10 more. Find the number of students who went.

10. Two pipes together fill a tank in 6 hours. The larger pipe alone takes 5 hours less than the smaller. Find the time each takes alone.

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Answer Key

1. 11 and 13
2. 7 and 8
3. $4x^2 - 15x - 4 = 0$ gives $x = 4$
4. $(x - 2)(x + 4) = 91$ gives $x = 9$ years
5. 12 cm by 7 cm
6. $x(x + 3) = 108$ gives $x = 9$; sides 9 cm and 12 cm
7. $x^2 + 15x - 4500 = 0$ gives 60 km/h
8. $x^2 - 1 = 24$ gives 5 km/h
9. $x^2 - 4x - 192 = 0$ gives 16 planned, so 12 went
10. $x^2 - 17x + 30 = 0$ gives 15 hours and 10 hours

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